

ZINC KRAS Hit Triage Report

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Objective

Prioritize computational KRAS hit candidates from a sampled ZINC lead-like screening library using the trained KRAS bioactivity model and the agentic triage workflow.

Hit Selection Criteria

- Trained-model inference required: True
- Minimum KRAS activity probability: 0.7
- Minimum ADMET score: 0.6
- Required toxicity label: Low risk
- Allowed recommendations: Advance, Advance with review

Screening Summary

- Total screened compounds: 5000
- Computational hit candidates passing filters: 4
- Reported candidates: 4

Top Hit Candidates

Rank	Compound	KRAS Probability	Overall	ADMET	Toxicity	Recommendation
1	72411788	0.816	0.836	0.826	Low risk	Advance
2	72408688	0.774	0.826	0.826	Low risk	Advance
3	72453647	0.709	0.812	0.837	Low risk	Advance with review
4	72422287	0.704	0.808	0.826	Low risk	Advance with review

Candidate Interpretations

1. 72411788

72411788 is prioritized because it shows a moderate trained-model KRAS activity signal (probability 0.816), acceptable ADMET profile (score 0.826), toxicity label Low risk, and agent recommendation Advance.

- Target-fit label: Predicted KRAS-active
- Inference mode: trained_model
- Mutant-selectivity hypothesis: Best hypothesis: SOS1
- Manufacturability: Complex synthesis with score 0.601
- SMILES: CCc1nc2c(c(N3CC[C@H](COC)C3)n1)CCN(CC(=O)O)CC2

2. 72408688

72408688 is prioritized because it shows a moderate trained-model KRAS activity signal (probability 0.774), acceptable ADMET profile (score 0.826), toxicity label Low risk, and agent recommendation Advance.

- Target-fit label: Predicted KRAS-active
- Inference mode: trained_model
- Mutant-selectivity hypothesis: Best hypothesis: SOS1
- Manufacturability: Complex synthesis with score 0.601
- SMILES: CCOCc1nc2c(c(N3CCC(O)CC3)n1)CCN(C(C)=O)CC2

3. 72453647

72453647 is prioritized because it shows a moderate trained-model KRAS activity signal (probability 0.709), acceptable ADMET profile (score 0.837), toxicity label Low risk, and agent recommendation Advance with review.

- Target-fit label: Predicted KRAS-active
- Inference mode: trained_model
- Mutant-selectivity hypothesis: Best hypothesis: SOS1
- Manufacturability: Complex synthesis with score 0.613
- SMILES: COCCNc1nc2c(c(N3CCC[C@@H](CO)C3)n1)CCNCC2

4. 72422287

72422287 is prioritized because it shows a moderate trained-model KRAS activity signal (probability 0.704), acceptable ADMET profile (score 0.826), toxicity label Low risk, and agent recommendation Advance with review.

- Target-fit label: Predicted KRAS-active
- Inference mode: trained_model
- Mutant-selectivity hypothesis: Best hypothesis: SOS1
- Manufacturability: Complex synthesis with score 0.601
- SMILES: CCNC(=O)[C@H]1CCN(c2nc(COCC)nc3c2CCNCC3)C1

Interpretation

These molecules should be treated as computational hit candidates, not experimentally confirmed KRAS inhibitors. The next scientific step is structure-based validation, such as docking against a relevant KRAS mutant structure, followed by deeper medicinal chemistry review and experimental assay planning.